QBITEL BRIDGE

Banking & Financial Services Security

Quantum-Safe Protection for Global Payment Infrastructure

Securing ISO 8583 | SWIFT/FedWire | COBOL Mainframes | FIX Trading

10,000+	<50ms	78%	PCI-DSS
TPS Encryption	End-to-End Latency	Autonomous Response	4.0 Ready Certified

enterprise@qbitel.com | https://bridge.qbitel.com

EXECUTIVE SUMMARY

Quantum-Safe Protection for Global Payment Infrastructure

The global banking system processes over $5 trillion in daily transactions across payment rails built on protocols designed decades before quantum computing existed. ISO 8583 card networks, SWIFT MT/MX wire transfers, COBOL mainframes, and FIX trading systems form the backbone of global finance - and every one carries quantum-vulnerable cryptography that nation-state adversaries are harvesting today for decryption tomorrow.

QBITEL Bridge is the only protocol-aware, quantum-safe security platform purpose-built for banking infrastructure. It delivers post-quantum encryption at **10,000+ transactions per second** with **sub-50ms latency**, integrates natively with HSMs from Thales Luna, AWS CloudHSM, and Futurex, and provides **78% autonomous incident response** without touching your core banking systems.

$6.2B	80%	<50ms	78%
Projected quantum-related banking losses by 2030	Payment volume on quantum-vulnerable rails	PQC encryption latency at 10K+ TPS	Autonomous incident response

* **$6.2 billion** in quantum-related banking losses projected by 2030 (NIST/McKinsey)
* **80%** of global payment volume traverses quantum-vulnerable legacy systems
* **DORA** (EU Digital Operational Resilience Act) ICT risk frameworks - mandatory January 2025
* **PCI-DSS 4.0** requires cryptographic agility and quantum-readiness assessments
* **Harvest-now-decrypt-later** attacks on SWIFT traffic confirmed by intelligence agencies

QBITEL Bridge enables your institution to achieve quantum-safe compliance, protect payment rails in production, and modernize security posture - all without disrupting the 15-20 year system lifecycles that define banking infrastructure.

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THE BANKING CRISIS NO CISO CAN IGNORE

Three existential threats to global payment infrastructure

Threat 1: Harvest-Now-Decrypt-Later on Payment Rails

Nation-state actors - confirmed by NSA, GCHQ, and ENISA advisories - are systematically harvesting encrypted SWIFT MT103 wire transfers, ISO 8583 card authorization flows, and FedWire/ACH batch files today. Their strategy: store the ciphertext, wait for quantum computers to mature (5-10 years), then decrypt trillions in historical transaction records.

> * *IBM, Google, and IonQ roadmaps target fault-tolerant quantum systems within 5-10 years.*
> *SWIFT regulatory archive requirements: 5-7 years. The harvest window is already open.*
> *Average cost of a major payment breach: $50 million - quantum decryption has no precedent.*

Threat 2: Mainframe Inter-System Unencrypted Communications

IBM z/OS mainframes running COBOL/CICS/DB2 process the majority of global banking transactions. Inter-LPAR communications, TN3270e terminal sessions, MQ messaging between CICS regions, and DB2 DRDA database connections often traverse unencrypted or weakly encrypted channels - even in 2026. Traditional network security tools cannot inspect these because they do not understand EBCDIC encoding, CICS transaction flows, or DB2 package authentication.

Threat 3: COBOL Legacy Protocol Vulnerabilities

The average age of COBOL applications in production banking systems is 45 years. Architected before TLS existed, before IPv6, and before cryptographic agility. QBITEL research identified three vulnerability categories in production banking environments:

* **Replay attacks** on fixed-format ISO 8583 messages lacking nonce or timestamp fields
* **MITM exposure** on TN3270e sessions between branch teller systems and mainframe CICS
* **Weak key derivation** in legacy DES/3DES implementations in payment processing COBOL modules

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QBITEL BRIDGE FOR BANKING

Protocol-aware | Hardware-backed | Quantum-safe

QBITEL Bridge is deployed as a transparent proxy layer between existing banking infrastructure components. No modifications to core banking applications, no changes to SWIFT connectivity, no alterations to payment card network integrations.

Component	Capability	Banking Protocol Coverage
Protocol Discovery Engine	2-4 hour full inventory	ISO 8583, ISO 20022, SWIFT MT/MX, FIX, TN3270e, ACH, FedWire, SEPA, CHIPS
PQC Cryptographic Engine	NIST FIPS 203/204/205	ML-KEM (Kyber), ML-DSA (Dilithium), SPHINCS+
HSM Integration Layer	PKCS#11 + native APIs	Thales Luna 7, AWS CloudHSM, Azure MHSM, Futurex, Utimaco
Autonomous Response Fabric	78% zero-touch response	Payment fraud, SWIFT anomaly, mainframe intrusion playbooks
Compliance Automation	Real-time evidence	PCI-DSS 4.0, DORA, Basel III/IV, SOX, GDPR, BCBS 239, SWIFT CSP

* *Deployment: On-premises appliance* | *Private cloud (VMware/OpenStack)* | *Public cloud (AWS/Azure/GCP) BYOK* | *Hybrid*

CAPABILITY 1: ISO 8583 / ISO 20022 PAYMENT RAIL PROTECTION

Field-level PQC encryption at 10,000+ TPS for card payment networks

ISO 8583 is the dominant message standard for card payment authorization, clearing, and settlement - used by Visa, Mastercard, Amex, and domestic card schemes globally. ISO 20022 is the next-generation XML-based standard adopted by SWIFT, TARGET2, CHIPS, and FedNow.

The problem: Both standards define message structure but not cryptographic protection. ISO 8583 messages typically use PIN block encryption for PIN fields only - the remaining 128 data elements including PANs, amounts, merchant codes, and authorization codes traverse in cleartext or with legacy RSA-1024/2048 transport encryption.

Protection Layer	QBITEL Bridge Capability	Standard Preserved
ISO 8583 Field Encryption	Field-level ML-KEM for PAN, expiry, CVV2, amount	Full bit-map integrity
Protocol-Aware Inspection	Parse bit maps at 10,000+ TPS without buffering	Zero message modification
ISO 20022 XML Signing	ML-DSA on pacs.008, camt.053, pain.001 message types	XSD schema compliance
PIN Block Modernization	TDES to AES-256 with PQC key wrapping (ISO 9564)	PIN block format preserved
Cryptographic Agility	Algorithm swap without application code change	Scheme format integrity maintained

10K+	<50ms	128	Zero
ISO 8583 TPS with full PQC active	End-to-end encryption latency	ISO 8583 fields protected selectively	Application code changes required

CAPABILITY 2: SWIFT / WIRE TRANSFER SECURITY

Post-quantum message signing for MT103, MT202, SWIFT MX, FedWire, CHIPS

SWIFT is the messaging backbone for international wire transfers, correspondent banking, and securities settlement. SWIFT MT103, MT202, and ISO 20022 MX messages carry instructions for trillions in daily interbank flows. SWIFT MX migration is underway and mandatory for most corridors.

> * *Bangladesh Bank 2016: $81M stolen via fraudulent MT103 insertions - no message-level signing.*
> *SWIFT regulatory archives carry RSA/AES encryption: quantum computers will break this.*
> *SWIFT CSP 2025 mandates controls but does not yet mandate post-quantum cryptography.*

SWIFT CSP Control	QBITEL Bridge Capability	Algorithm
MT/MX Message Authentication	ML-DSA signature on every outbound SWIFT message	FIPS 204 ML-DSA
BIC-to-BIC Key Exchange	ML-KEM replacing RSA-2048 key transport	FIPS 203 ML-KEM
FedWire/Fedline Protection	PQC-wrapped Fedline Advantage connections	AES-256-GCM + ML-KEM
CHIPS Authentication	PQC key wrapping for CHIPS participants	ML-KEM + ML-DSA
SWIFT Anomaly Detection	ML model trained on 50M+ SWIFT message patterns	Real-time inference
CSP 1.1 Environment	Protocol-aware network segmentation	Zero-trust micro-segmentation

CAPABILITY 3: COBOL / MAINFRAME LEGACY SHIELD

Zero code changes. TN3270e PQC proxy, CICS signing, DB2 DRDA upgrade.

IBM z/OS mainframes running COBOL/CICS/DB2 process an estimated **95% of ATM transactions** and **80% of in-person card swipes** globally. With 15-20 year deployment lifecycles, these systems cannot be replaced without multi-year transformation programs. QBITEL Bridge brings quantum-safe security to the mainframe without touching a single line of COBOL.

Component	Zero-Downtime	Performance Impact	Code Change
TN3270e Sessions	YES	<5ms per session	None
CICS Transactions	YES	<2ms per transaction	None
DB2 DRDA Connections	YES	<8ms per query	None
IBM MQ Messages	YES	<3ms per message	None
VSAM File Encryption	Scheduled window	<1% MIPS overhead	None
IMS DL/I Calls	YES	<4ms per call	None
AS/400 (IBM i)	YES	<6ms per session	None

CAPABILITY 4: TRADING PROTOCOL SECURITY (FIX / FPML)

ML-DSA message signing for FIX sessions at <50 microseconds latency

Electronic trading relies on FIX (Financial Information eXchange) protocol for order routing, execution confirmation, and market data between buy-side, sell-side, ECNs, and exchanges. FIX 4.2 through FIX 5.0/FIXT 1.1 carries equity, fixed income, FX, and derivatives orders globally. FpML covers OTC derivatives confirmation under EMIR/Dodd-Frank.

FIX/FpML Protection	Technical Detail
FIX Session Authentication	Replace Tag 96 password with ML-DSA signing on every message
Order Message Signing	NewOrderSingle (35=D), OrderCancel (35=F), ExecutionReport (35=8)
Replay Prevention	Bind ML-DSA signatures to MsgSeqNum - cryptographic replay block
FpML Encryption	ML-KEM + AES-256-GCM for OTC confirmation payloads (EMIR compliant)
Market Data Integrity	Sign MarketDataSnapshotFullRefresh (35=W) from exchange feeds
Latency Target	<50 microseconds added latency - DPDK-accelerated processing

* *Red team finding: FIX sequence number replay allows order book manipulation.*
QBITEL Bridge: ML-DSA signing deployed in 48 hours, <50us latency impact, zero code changes.

CAPABILITIES 5-7: CLOUD SECURITY | COMPLIANCE | FRAUD ANALYTICS

Enterprise-wide PQC coverage across cloud, compliance, and fraud prevention

Capability 5: Cloud Migration Security

QBITEL Bridge enables BYOK post-quantum keys into AWS CloudHSM, Azure Managed HSM, and GCP Cloud HSM. ML-KEM outer-layer wraps all cloud-native encryption operations. Multi-cloud key synchronization with GDPR/DORA data residency enforcement.

Provider	HSM Product	Integration	PQC Support
AWS	CloudHSM (Luna HSM 7)	PKCS#11 + JCA/JCE	ML-KEM, ML-DSA via QBITEL proxy
Azure	Managed HSM (Marvell)	Azure MHSM REST API	ML-KEM, ML-DSA via QBITEL proxy
GCP	Cloud HSM (Marvell)	PKCS#11 + KMS API	ML-KEM, ML-DSA via QBITEL proxy
On-prem	Thales Luna Network HSM 7	PKCS#11 + REST	Native + QBITEL extensions
On-prem	Futurex Vectera Plus	PKCS#11 + FXAPI	Native + QBITEL extensions

Capability 6: Autonomous Compliance - PCI-DSS 4.0 / DORA / Basel III

Automates evidence collection for PCI-DSS 4.0 (Requirements 3, 4, 6, 8, 10, 12), DORA (Articles 9, 10, 11, 17, 26, 28), Basel III/IV operational risk data, SOX ITGC controls, GDPR data-by-default encryption, BCBS 239 data lineage, SWIFT CSP 2025, and NY DFS Part 500.

Capability 7: Real-Time Fraud Analytics

Protocol-native feature extraction from ISO 8583 messages: 200+ features including BIN geography, merchant category cross-reference, amount velocity, and field consistency scoring - all at sub-50ms inference integrated directly into the authorization path. SWIFT MT103 anomaly detection covers beneficiary manipulation, amount rounding (BEC hallmark), and off-hours timing patterns. Federated learning trains models without sharing raw transaction data (GDPR compliant).

COMPLIANCE COVERAGE

PCI-DSS 4.0 | DORA | Basel III/IV | SOX | GDPR | BCBS 239 | SWIFT CSP

Regulation	QBITEL Bridge Coverage	Evidence Generated
PCI-DSS 4.0	Requirements 3, 4, 6, 8, 10, 12	SAQ-D, ROC evidence packages
DORA (EU) 2022/2554	Articles 9, 10, 11, 17, 26, 28	ICT risk register, incident reports
Basel III/IV	Operational risk data collection	AMA data feeds, KRI dashboards
SOX Section 404	IT general controls (ITGC)	ITGC control evidence
GDPR Articles 25/32	Data-by-default encryption	ROPA entries, DPA evidence
BCBS 239	Risk data aggregation reporting	Data lineage maps
SWIFT CSP 2025	Mandatory + advisory controls	CSP attestation package
NIST FIPS 140-3	Level 3 HSM validation	CMVP certificates
ISO 27001:2022	Annex A cryptographic controls	ISMS evidence
NY DFS Part 500	Encryption and CISO reporting	Part 500 attestation

INTEGRATION ECOSYSTEM

Core banking | Payment infrastructure | Security stack

Core Banking Systems

Vendor	Product	Integration Method	Deploy Time
Temenos	T24 / Transact	REST API + message broker	2-4 weeks
Finastra	Fusion / Kondor	SWIFT gateway integration	3-5 weeks
FIS	Modern Banking Platform	ISO 20022 proxy	2-3 weeks
Fiserv	DNA / Premier	JDBC/ODBC intercept	1-2 weeks
Oracle	FLEXCUBE	MQ + REST integration	3-4 weeks
SAP	SAP Banking Services	RFC/BAPI interception	2-4 weeks
Infosys	Finacle	REST + SWIFT gateway	2-3 weeks

Payment Infrastructure

System	Protocol	Notes
Visa DPS / VisaNet	ISO 8583	Transparent proxy - zero application change
Mastercard	ISO 8583 / ISO 20022	Field-level encryption, scheme format preserved
SWIFT Alliance	MT/MX (ISO 20022)	SWIFT API Gateway, CSP-aligned
FedNow	ISO 20022	Real-time PQC on Fed connectivity layer
FedWire	Fedline protocol	FIPS 140-3 compliant TLS + PQC wrapper
CHIPS	Proprietary	PQC key wrapping for CHIPS participants
SEPA (EPC)	ISO 20022 SCT/SCT Inst	GDPR-compliant cross-border PQC
ACH / NACHA	NACHA format	File-level encryption, batch + real-time

DEPLOYMENT TIMELINE

4-phase deployment: 16 weeks from passive tap to full enterprise PQC

Phase 1: Protocol Discovery & Risk Assessment — Weeks 1-2

* Deploy passive protocol tap on SPAN/mirror ports - zero production impact
* Automated discovery: ISO 8583, SWIFT, FIX, TN3270e, and proprietary protocols
* Cryptographic inventory: identify all RSA, ECC, DES/3DES deployments
* Quantum risk scoring: rank protocols by breach impact and vulnerability timeline
* Deliverable: Protocol Risk Register with prioritized remediation roadmap

Phase 2: Infrastructure Readiness & HSM Provisioning — Weeks 3-5

* HSM cluster provisioning: Thales Luna / AWS CloudHSM / Azure MHSM
* Network segmentation: PCI cardholder data environment isolation
* QBITEL Bridge appliance deployment (physical or virtual)
* Integration testing with core banking system (read-only / passive mode)
* Deliverable: QBITEL Bridge operational in passive monitoring mode

Phase 3: Payment Rail Protection — Weeks 6-10

* ISO 8583 field-level encryption activation (non-business-hours cutover)
* SWIFT MT/MX PQC wrapping activation - zero SWIFT downtime
* FedWire/ACH session protection and performance validation
* 10,000 TPS load test + <50ms latency verification
* Deliverable: Payment rails protected with PQC; fraud analytics baseline operational

Phase 4: Mainframe, Legacy & Full Compliance — Weeks 11-16

* TN3270e PQC proxy activation for branch teller network
* COBOL/CICS transaction signing and DB2 DRDA PQC upgrade
* FIX/FpML trading protocol security activation
* DORA ICT risk register + PCI-DSS 4.0 evidence package completion
* Go-live sign-off, SOC integration, runbook handover

PERFORMANCE SPECIFICATIONS

Validated under sustained load with ML-KEM and ML-DSA active

Metric	Specification	Test Condition
ISO 8583 throughput	10,000+ TPS sustained	60-minute load test, ML-KEM active
SWIFT MT103 latency	<8ms added latency	ML-DSA signing, full HSM custody
FIX message latency	<50 microseconds added	FIX 5.0 NewOrderSingle with signing
TN3270e session setup	<5ms per session	2,000 concurrent sessions
Protocol discovery	2-4 hours complete	100+ protocol types on 10Gbps link
Autonomous response rate	78% zero-touch	Banking-tuned incident playbooks
System availability	99.999% (5 nines)	Active-active HA cluster
HSM operations	20,000 ops/sec	Thales Luna Network HSM 7
Key rotation	Zero-downtime	Automated with HSM custody
Failover time	<30 seconds	Active-passive with synchronous replication
Audit log throughput	100,000 events/sec	Signed, tamper-evident logs
Compliance reporting	Real-time	Continuous evidence collection

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COMPETITIVE DIFFERENTIATION

Why QBITEL Bridge is the only banking-native PQC platform

vs. Traditional HSM Vendors (Thales, Utimaco, Futurex)

HSMs provide cryptographic operations but have zero protocol intelligence. An HSM cannot parse ISO 8583 bit maps, inspect SWIFT message fields, or identify COBOL data structures. QBITEL Bridge uses HSMs as the cryptographic backend and adds protocol intelligence, autonomous response, and compliance automation on top.

> * *QBITEL: Protocol-aware + HSM-backed = the only combination addressing both protocol and cryptographic security.*

vs. Cloud-Native Encryption (AWS KMS, Azure Key Vault, GCP KMS)

Cloud KMS services use RSA-2048 and ECC P-256 for key transport - both quantum-vulnerable. Post-quantum support in cloud KMS is in beta with limited algorithm coverage. Cloud KMS cannot protect on-premises mainframes, SWIFT connectivity, or ISO 8583 card networks.

> * *QBITEL: Cloud-agnostic, mainframe-capable, fully PQC-compliant today - not on a roadmap.*

vs. Network Security Platforms (Palo Alto, Fortinet, Cisco)

Next-generation firewalls operate on IP/TCP layers and cannot decrypt banking-specific protocols without breaking transaction flows. They provide TLS inspection for web traffic but have no concept of ISO 8583 field structures or SWIFT message types.

> * *QBITEL: Banking-protocol-native inspection at layers 5-7 - not just network-layer perimeter.*

vs. Payment Security Specialists (Voltage, P2PE vendors)

Point-to-point encryption vendors focus on PAN protection at the point of sale. They do not address SWIFT security, mainframe communications, trading protocol integrity, or post-quantum key exchange for wire transfer systems.

> * *QBITEL: Complete ecosystem - ISO 8583, SWIFT, FedWire, COBOL mainframes, FIX trading, cloud.*

CUSTOMER SCENARIOS

Real deployments across Tier-1 global bank, regional bank, and investment bank

Scenario 1 > Tier-1 Global Bank - DORA Compliance Under Time Pressure

Challenge: EU universal bank with 90-day deadline to demonstrate DORA remediation to regulator.
Discovery: 340 protocol flows found in 72 hours - 23 SWIFT paths, 4 card networks, 12 TN3270e sessions.
Deployment: ML-KEM activated on SWIFT connections in Week 3 - zero SWIFT connectivity interruption.
Compliance: DORA ICT risk register auto-populated; incident engine connected to Splunk in Week 4.
Outcome: DORA evidence package delivered 45 days ahead of regulatory deadline. ECB-accepted.
Metrics: 340 protocols in 72h | DORA package in 45 days | Zero SWIFT downtime

Scenario 2 > Regional US Bank - PCI-DSS 4.0 Quantum Readiness

Challenge: $15B asset community bank - QSA finding: RSA-1024 in ISO 8583 acquirer connection.
Discovery: Passive tap on card network connections revealed legacy RSA-1024 key exchange.
Deployment: ML-KEM replacement activated in 48 hours with zero card network downtime.
Compliance: PCI-DSS 4.0 evidence auto-generated for Requirements 3, 4, 8, and 10.
Outcome: RSA-1024 critical finding closed. QSA ROC evidence package auto-generated. Zero findings.
Metrics: Critical finding remediated in 48h | PCI-DSS ROC in 30 days | QSA findings closed

Scenario 3 > Investment Bank - FIX Trading Protocol Security

Challenge: Red team critical finding - FIX 4.2 sequence number replay allows order book manipulation.
Discovery: FIX session monitoring activated in 4 hours via passive tap. Replay attack confirmed.
Deployment: ML-DSA signing on all FIX sessions in 48 hours. Zero trading system code changes.
Security: Sequence number binding to ML-DSA signature prevents replay cryptographically.
Outcome: Critical finding remediated. <50 microseconds latency impact. EMIR FpML compliance.
Metrics: 48h remediation | <50us latency | Zero code changes | EMIR FpML compliance

NEXT STEPS

Protocol discovery to full compliance in 16 weeks

1	**Protocol Discovery Session** Deploy passive tap. Deliver complete banking protocol inventory and quantum risk assessment within 48 hours. Zero production impact, no agents, no code changes.
2	**Compliance Gap Assessment** Map current cryptographic posture against PCI-DSS 4.0, DORA, and SWIFT CSP 2025. Identify highest-priority gaps with remediation roadmap.
3	**Technical Deep-Dive** 90-minute session for CISO, Head of Payments, and Head of Architecture. Protocol-specific deployment design for your ISO 8583, SWIFT, and mainframe topology.
4	**30-Day Proof of Value** Production POV: protocol discovery, PQC activation on one payment rail, PCI-DSS 4.0 compliance gap report, DORA ICT risk register seed, performance benchmarking. No long-term commitment. Most clients achieve ROI within Q1.

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CONTACT QBITEL BANKING PRACTICE

enterprise@qbitel.com | https://bridge.qbitel.com

NIST	PCI-DSS	SWIFT	SOC 2
FIPS 140-3 Level 3	QSA Partner Program	Service Bureau Partner	Type II Certified

Enterprise Inquiries	Platform Portal	Banking Practice
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* *QBITEL Bridge - Quantum-Safe. Protocol-Aware. Banking-Ready.*
Protecting ISO 8583 | *SWIFT MT/MX* | *COBOL Mainframes* | *FIX Trading* | *Real-Time Payments*